

Multiple Choice

1. **Answer:** D
Options: A) 3; B) 4; C) 7; D) 15
Note: Correct option: D - 15
2. **Answer:** B
Options: A) $O(1)$; B) $O(N)$; C) $O(\log N)$; D) $O(N^2)$
Note: Correct option: B - $O(N)$
3. **Answer:** D
Options: A) To save computing time during testing; B) To save space for storing the Decision Tree; C) To make the training set error smaller; D) To avoid overfitting the training set
Note: Correct option: D - To avoid overfitting the training set
4. **Answer:** D
Options: A) True, True; B) False, False; C) True, False; D) False, True
Note: Correct option: D - False, True
5. **Answer:** A
Options: A) True, True; B) False, False; C) True, False; D) False, True
Note: Correct option: A - True, True

True / False

6. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'polynomial in D'.
7. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'True, True'.
8. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'Decrease variance'.
9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** `def __init__(self, a, b): | def max_chain_length(arr, n): | return max`

Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** `def max_sum_pair_diff_less_than_K(arr, N, K): | return dp[N - 1]`

Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key